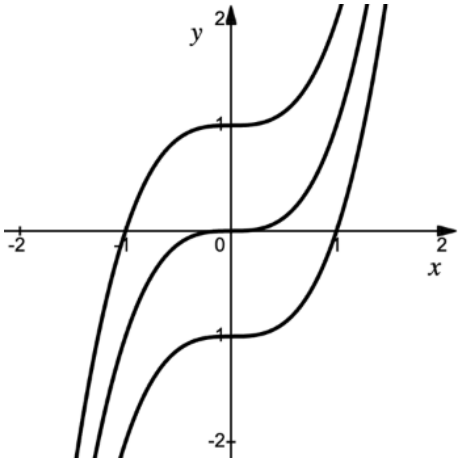
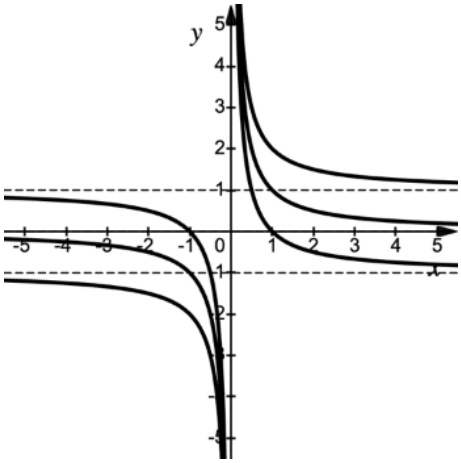
1)



2)



3) a)

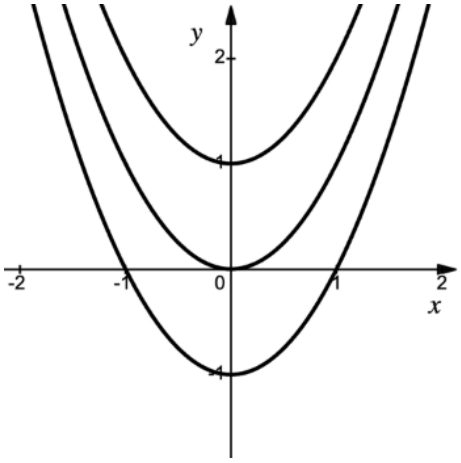
The gradient is zero when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \Rightarrow 2x = 0 \Rightarrow x = 0$

3) b)

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x$

$\Rightarrow y = x^2 + c$

3) c)



3) d)

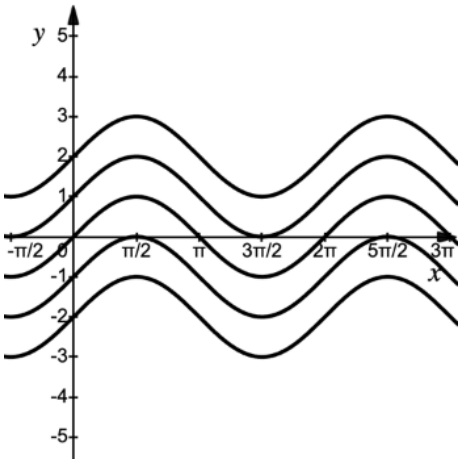
The family of solutions are vertical translations of $y = x^2$

4) a)

$\frac{dy}{dx} = \cos x$

$\Rightarrow y = \sin x + c$

4) b)



5) a)

$\frac{\mathrm{d}y}{\mathrm{d}x} = 3y$

$\Rightarrow \int \frac{1}{y}\mathrm{d}y = \int 3\,\mathrm{d}x$

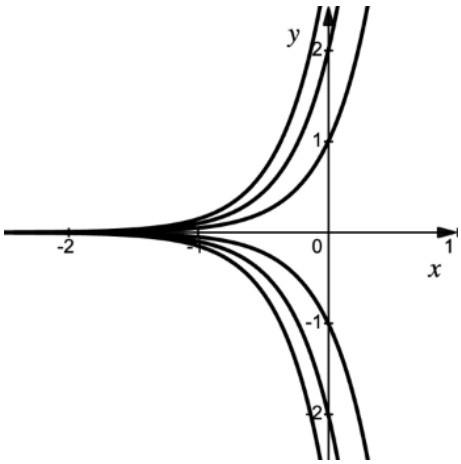
$\Rightarrow \ln|y| = 3x + c$

$\Rightarrow y = \pm \mathrm{e}^{3x+c}$

$\Rightarrow y = \pm \mathrm{e}^{3x}\mathrm{e}^{c}$

$\Rightarrow y = A\mathrm{e}^{3x}$

5) b)



5) c)

As $x \to -\infty$, $y \to 0$ (the x-axis is a horizontal asymptote)

6) a)

When $y = 2$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 2 - 2 = 0$ and so is a constant solution to the differential equation.

6) b)

$\frac{\mathrm{d}y}{\mathrm{d}x} = y - 2$

$\Rightarrow \int \frac{1}{y-2}\mathrm{d}y = \int \mathrm{d}x$

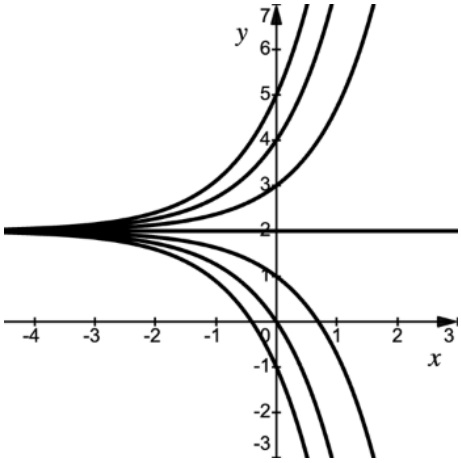
$\Rightarrow \ln|y - 2| = x + c$

$\Rightarrow y - 2 = \pm \mathrm{e}^{x+c}$

$\Rightarrow y = 2 \pm \mathrm{e}^{x}\mathrm{e}^{c}$

$\Rightarrow y = 2 + A\mathrm{e}^{x}$

6) c)



7) a)

$\frac{\mathrm{d}y}{\mathrm{d}x} = -2y$

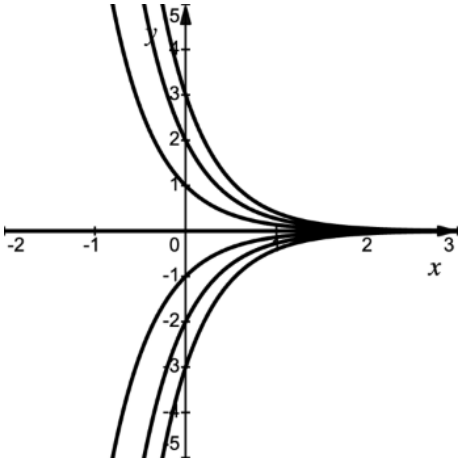
$\Rightarrow \int \frac{1}{y}\mathrm{d}y = \int -2\,\mathrm{d}x$

$\Rightarrow \ln|y| = -2x + c$

$\Rightarrow y = \pm \mathrm{e}^{-2x+c}$

$\Rightarrow y = \pm \mathrm{e}^{-2x}\mathrm{e}^{c}$

$\Rightarrow y = A\mathrm{e}^{-2x}$



7) b)

$\frac{\mathrm{d}y}{\mathrm{d}x} = 1 - y$

$\Rightarrow \int \frac{1}{1-y}\mathrm{d}y = \int \mathrm{d}x$

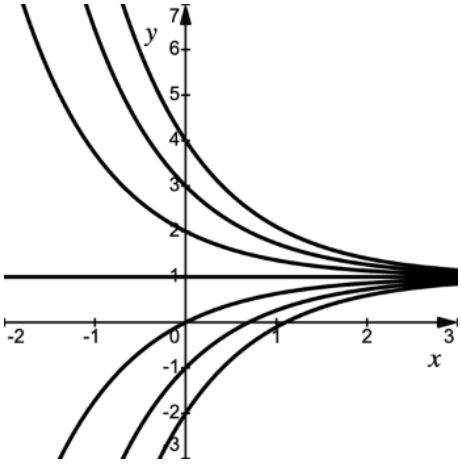
$\Rightarrow -\ln|1-y| = x + c$

$\Rightarrow \ln|1-y| = k - x$

$\Rightarrow 1 - y = \pm \mathrm{e}^{k-x}$

$\Rightarrow y = 1 \pm \mathrm{e}^{k}\mathrm{e}^{-x}$

$\Rightarrow y = 1 + A\mathrm{e}^{-x}$



7) c)

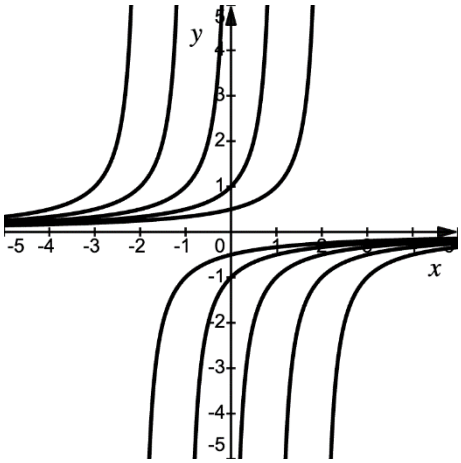
$\frac{\mathrm{d}y}{\mathrm{d}x} = y^2$

$\Rightarrow \int \frac{1}{y^2}\mathrm{d}y = \int \mathrm{d}x$

$\Rightarrow -\frac{1}{y} = x + c$

$\Rightarrow \frac{1}{y} = A - x$

$\Rightarrow y = \frac{1}{A-x}$



7) d)

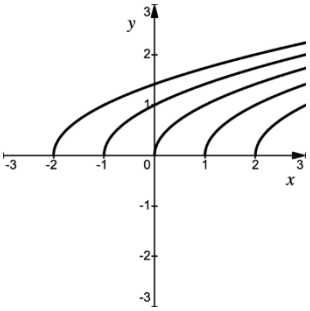
$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2y}$

$\Rightarrow \int 2y \, \mathrm{d}y = \int \mathrm{d}x$

$\Rightarrow y^2 = x + c$

$\Rightarrow y = \pm\sqrt{x + c}$

Given that $y > 0$, $y = \sqrt{x + c}$



8) a)

When $y = 0$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 0(1 - 0) = 0$ and when $y = 1$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 1(1 - 1) = 0$

So $y = 0$ and $y = 1$ are constant solutions to the differential equation.

8) b)

$\frac{\mathrm{d}y}{\mathrm{d}x} = y(1-y)$

$\Rightarrow \int \frac{1}{y(1-y)}\mathrm{d}y = \int \mathrm{d}x$

Use Partial Fractions:

$\frac{1}{y(1-y)} \equiv \frac{A}{y} + \frac{B}{1-y}$

$\Rightarrow 1 \equiv A(1-y) + By$

Let $y = 0$: $1 = A$

Let $y = 1$: $1 = B$

$\therefore \frac{1}{y(1-y)} \equiv \frac{1}{y} + \frac{1}{1-y}$

$\Rightarrow \int \left(\frac{1}{y} + \frac{1}{1-y}\right)\mathrm{d}y = \int \mathrm{d}x$

$\Rightarrow \ln|y| - \ln|1-y| = x + c$

$\Rightarrow \ln\left|\frac{y}{1-y}\right| = x + c$

$\Rightarrow \frac{y}{1-y} = \pm \mathrm{e}^{x+c}$

$\Rightarrow \frac{y}{1-y} = \pm \mathrm{e}^{x}\mathrm{e}^{c}$

$\Rightarrow \frac{y}{1-y} = A\mathrm{e}^{x}$

$\Rightarrow y = A\mathrm{e}^{x} - A\mathrm{e}^{x}y$

$\Rightarrow y + A\mathrm{e}^{x}y = A\mathrm{e}^{x}$

$\Rightarrow (1 + \mathrm{Ae}^{x})y = A\mathrm{e}^{x}$

$\Rightarrow y = \frac{A\mathrm{e}^{x}}{1+A\mathrm{e}^{x}}$

8) c)

In part a), we found $y = 0$ and $y = 1$ are constant solutions of the differential equation.

If $y = 0$:

$0 = \frac{A\mathrm{e}^{x}}{1+A\mathrm{e}^{x}} \Rightarrow A = 0$

If $y = 1$:

$1 = \frac{A\mathrm{e}^{x}}{1+A\mathrm{e}^{x}} \Rightarrow 1 + A\mathrm{e}^{x} = A\mathrm{e}^{x} \Rightarrow 1 = 0$

So $y = 1$ does not appear in the solution found in part b), but $y = 0$ does.

When we separate the variables from $\frac{\mathrm{d}y}{\mathrm{d}x} = y(1-y)$ to $\int \frac{1}{y(1-y)}\mathrm{d}y = \int \mathrm{d}x$ we are assuming $y \neq 0$ and $y \neq 1$

8) d)

$$y = \frac{Ae^x}{1+Ae^x}$$
$$= \frac{\frac{Ae^x}{Ae^x}}{\frac{1}{Ae^x}+\frac{Ae^x}{Ae^x}}$$
$$= \frac{1}{\frac{1}{A}e^{-x}+1}$$
$$= \frac{1}{1+Be^{-x}}$$

8) e)

In part a), we found $y = 0$ and $y = 1$ are constant solutions of the differential equation.

If $y = 0$:

$0 = \frac{1}{1+Be^{-x}} \Rightarrow 0 = 1$

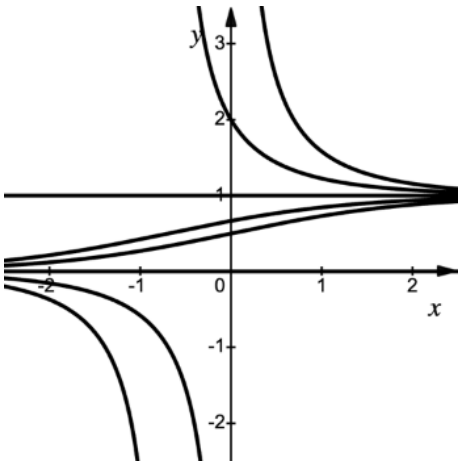
If $y = 1$:

$1 = \frac{1}{1+Be^{-x}} \Rightarrow 1 + Be^{-x} = 1 \Rightarrow B = 0$

So $y = 0$ does not appear in the solution found in part d), but $y = 1$ does.

When we separate the variables from $\frac{dy}{dx} = y(1-y)$ to $\int \frac{1}{y(1-y)} dy = \int dx$ we are assuming $y \neq 0$ and $y \neq 1$

8) f)



9) a)

Trial:

$y = k$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\therefore 0 + k = 3 \Rightarrow k = 3$

So $y = 3$ is a constant solution of the differential equation.

9) b)

$\frac{\mathrm{d}y}{\mathrm{d}x} + y = 3$

Integrating Factor

$= \mathrm{e}^{\int 1 \mathrm{d}x}$

$= \mathrm{e}^{x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} + y = 3$

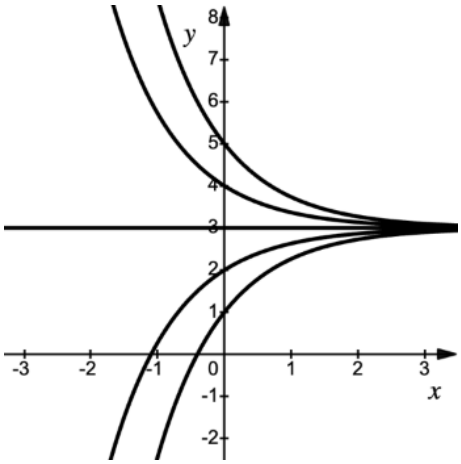
$\Rightarrow \mathrm{e}^{x} \frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{e}^{x} y = 3\mathrm{e}^{x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{x} y) = 3\mathrm{e}^{x}$

$\Rightarrow \mathrm{e}^{x} y = 3\mathrm{e}^{x} + c$

$\Rightarrow y = 3 + c\mathrm{e}^{-x}$

9) c)



10) a)

Trial:

$y = k$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\therefore 0 - k = 2 \Rightarrow k = -2$

So $y = -2$ is a constant solution of the differential equation.

10) b)

$\frac{\mathrm{d}y}{\mathrm{d}x} - y = 2$

Integrating Factor

$= \mathrm{e}^{\int -1 \mathrm{d}x}$

$= \mathrm{e}^{-x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} - y = 2$

$\Rightarrow \mathrm{e}^{-x}\frac{\mathrm{d}y}{\mathrm{d}x} - \mathrm{e}^{-x}y = 2\mathrm{e}^{-x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{-x}y) = 2\mathrm{e}^{-x}$

$\Rightarrow \mathrm{e}^{-x}y = -2\mathrm{e}^{-x} + c$

$\Rightarrow y = -2 + c\mathrm{e}^{x}$

10) c)

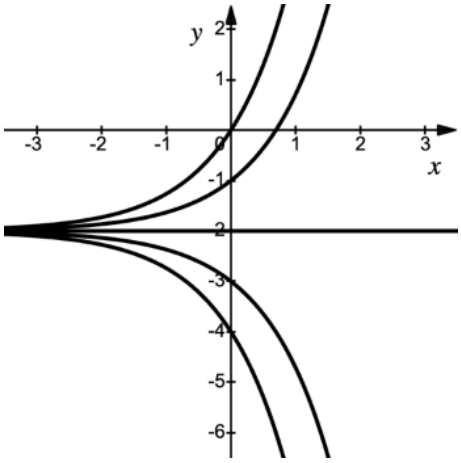
As $x \to \infty$, $\mathrm{e}^{x} \to \infty$

So if $c > 0$, then $y \to \infty$

If $c = 0$, then $y = -2$

If $c < 0$, then $y \to -\infty$

10) d)



11) a)

Trial:

$y = k$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\therefore 0 + 2k = 6 \Rightarrow k = 3$

So $y = 3$ is a constant solution of the differential equation.

11) b)

$\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 6$

Integrating Factor

$= \mathrm{e}^{\int 2\mathrm{d}x}$

$= \mathrm{e}^{2x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 6$

$\Rightarrow \mathrm{e}^{2x}\frac{\mathrm{d}y}{\mathrm{d}x} + 2\mathrm{e}^{2x}y = 6\mathrm{e}^{2x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{2x}y) = 6\mathrm{e}^{2x}$

$\Rightarrow \mathrm{e}^{2x}y = 3\mathrm{e}^{2x} + c$

$\Rightarrow y = 3 + c\mathrm{e}^{-2x}$

11) c)

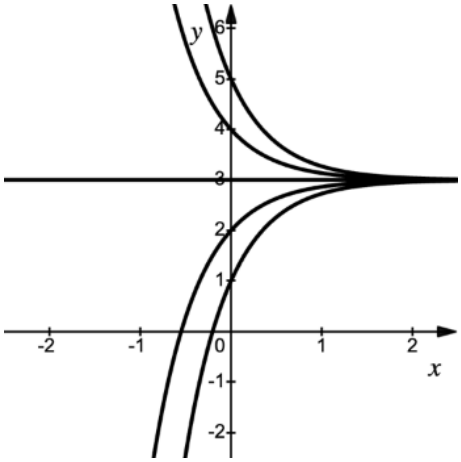
As $x \to \infty$, $e^{-2x} \to 0$

So if $c > 0$, then $y \to 3$

If $c = 0$, then $y = 3$

If $c < 0$, then $y \to 3$

11) d)



12) a)

Trial:

$y = k$

$\Rightarrow \frac{dy}{dx} = 0$

$\therefore 0 + 2x \times k = 4x \Rightarrow k = 2$

So $y = 2$ is a constant solution of the differential equation.

$\frac{dy}{dx} + 2xy = 4x$

Integrating Factor

$= e^{\int 2x dx}$

$= e^{x^2}$

$\frac{dy}{dx} + 2xy = 4x$

$\Rightarrow e^{x^2}\frac{dy}{dx} + 2xe^{x^2}y = 4xe^{x^2}$

$\Rightarrow \frac{d}{dx}\left(e^{x^2}y\right) = 4xe^{x^2}$

$\Rightarrow e^{x^2}y = 2e^{x^2} + c$

$\Rightarrow y = 2 + ce^{-x^2}$

12) b)

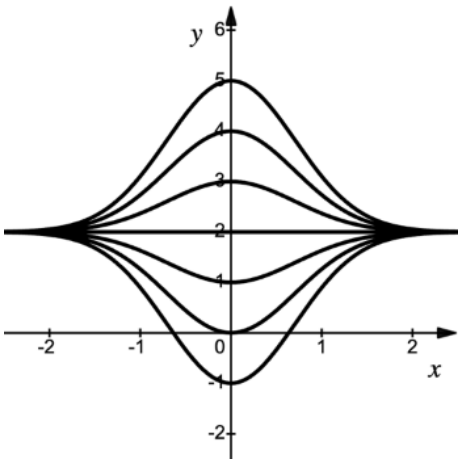
As $x \to \infty$, $e^{-x^2} \to 0$

So every solution curve tends to $y = 2$ as x tends to infinity

As $x \to -\infty$, $e^{-x^2} \to 0$

So every solution curve tends to $y = 2$ as x tends to negative infinity

12) c)



13) a)

$\frac{\mathrm{d}y}{\mathrm{d}x} + y = x + 1$

Integrating Factor

$= \mathrm{e}^{\int 1 \mathrm{d}x}$

$= \mathrm{e}^{x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} + y = x + 1$

$\Rightarrow \mathrm{e}^{x}\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{e}^{x}y = (x+1)\mathrm{e}^{x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{x}y) = (x+1)\mathrm{e}^{x}$

Use Integration by Parts:

$u = x + 1$ $\qquad$ $\frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{x}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 1$ $\qquad$ $v = \mathrm{e}^{x}$

$\therefore \int (x+1)\mathrm{e}^{x} \ln x \, \mathrm{d}x = (x+1)\mathrm{e}^{x} - \int \mathrm{e}^{x} \, \mathrm{d}x$

$= x\mathrm{e}^{x} + \mathrm{e}^{x} - \mathrm{e}^{x} + c$

$= x\mathrm{e}^{x} + c$

$\therefore \mathrm{e}^{x}y = x\mathrm{e}^{x} + c$

$\Rightarrow y = x + c\mathrm{e}^{-x}$

13) b)

$y = x \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 1$

So as $1 + x = x + 1$, $y = x$ is a solution of the differential equation.

13) c)

As $x \to \infty$, $\mathrm{e}^{-x} \to 0$

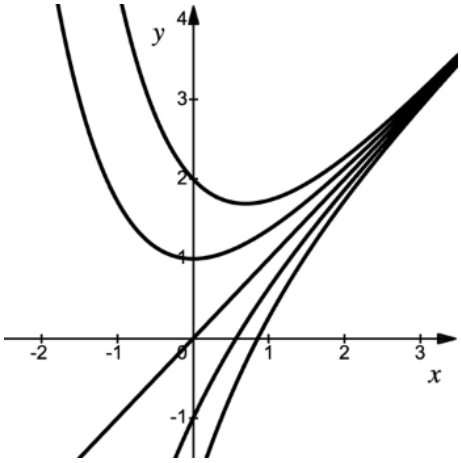
So if $c > 0$, then $y \to x$

If $c = 0$, then $y = x$

If $c < 0$, then $y \to x$

So every solution curve approaches the line $y = x$

13) d)



13) e)

Let one solution curve have equation $y = x + c_1 e^{-x}$ and let another solution curve have equation $y = x + c_2 e^{-x}$, where $c_1 \neq c_2$

If they were to intersect, then:

$x + c_1 e^{-x} = x + c_2 e^{-x}$

$\Rightarrow c_1 e^{-x} = c_2 e^{-x}$

$\Rightarrow c_1 = c_2$

But $c_1 \neq c_2$ so the solution curves cannot intersect.

14) a)

$\frac{\mathrm{d}y}{\mathrm{d}x} - y = x$

Integrating Factor

$= \mathrm{e}^{\int -1 \mathrm{d}x}$

$= \mathrm{e}^{-x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} - y = x$

$\Rightarrow \mathrm{e}^{-x}\frac{\mathrm{d}y}{\mathrm{d}x} - \mathrm{e}^{-x}y = x\mathrm{e}^{-x}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{-x}y) = x\mathrm{e}^{-x}$

Use Integration by Parts:

$u = x$ $\qquad$ $\frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{-x}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 1$ $\qquad$ $v = -\mathrm{e}^{-x}$

$\therefore \int x\mathrm{e}^{-x}\,\mathrm{d}x = -x\mathrm{e}^{-x} + \int \mathrm{e}^{-x}\,\mathrm{d}x$

$= -x\mathrm{e}^{-x} - \mathrm{e}^{-x} + c$

$\therefore \mathrm{e}^{-x}y = -x\mathrm{e}^{-x} - \mathrm{e}^{-x} + c$

$\Rightarrow y = -x - 1 + c\mathrm{e}^{x}$

14) b)

$y = -x - 1 \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -1$

So as $-1 - (-x - 1) = x$, $y = -x - 1$ is a solution of the differential equation.

14) c)

As $x \to -\infty$, $\mathrm{e}^{x} \to 0$

So if $c > 0$, then $y \to -x - 1$

If $c = 0$, then $y = -x - 1$

If $c < 0$, then $y \to -x - 1$

So every solution curve approaches the line $y = -x - 1$

14) d)

As $x \to \infty$, $e^x \to \infty$

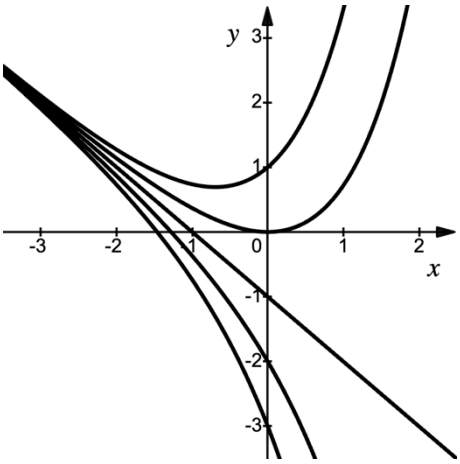
So if $c > 0$, then $y \to \infty$

If $c = 0$, then $y = -x - 1$

If $c < 0$, then $y \to -\infty$

So every solution curve diverges from the line $y = -x - 1$, unless $c = 0$

14) e)



15) a)

$\frac{\mathrm{d}y}{\mathrm{d}x} + y = \sin x + \cos x$

Integrating Factor

$= \mathrm{e}^{\int 1 \mathrm{d}x}$

$= \mathrm{e}^{x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} + y = \sin x + \cos x$

$\Rightarrow \mathrm{e}^{x}\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{e}^{x}y = \mathrm{e}^{x}(\sin x + \cos x)$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{x}y) = \mathrm{e}^{x}(\sin x + \cos x)$

Let $I = \int \mathrm{e}^{x}(\sin x + \cos x)\,\mathrm{d}x$

Use Integration by Parts:

$u = \sin x + \cos x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{x}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \cos x - \sin x \qquad v = \mathrm{e}^{x}$

$\therefore I = \mathrm{e}^{x}(\sin x + \cos x) - \int \mathrm{e}^{x}(\cos x - \sin x)\,\mathrm{d}x$

Use Integration by Parts again:

$u = \cos x - \sin x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{x}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = -\sin x - \cos x \qquad v = \mathrm{e}^{x}$

$\therefore I = \mathrm{e}^{x}(\sin x + \cos x) - (\mathrm{e}^{x}(\cos x - \sin x) - \int \mathrm{e}^{x}(-\sin x - \cos x)\,\mathrm{d}x)$

$\Rightarrow I = \mathrm{e}^{x}(\sin x + \cos x) - (\mathrm{e}^{x}(\cos x - \sin x) + \int \mathrm{e}^{x}(\sin x + \cos x)\,\mathrm{d}x)$

$\Rightarrow I = \mathrm{e}^{x}(\sin x + \cos x) - \mathrm{e}^{x}(\cos x - \sin x) - \int \mathrm{e}^{x}(\sin x + \cos x)\,\mathrm{d}x$

$\Rightarrow I = \mathrm{e}^{x}\sin x + \mathrm{e}^{x}\cos x - \mathrm{e}^{x}\cos x + \mathrm{e}^{x}\sin x - I$

$\Rightarrow 2I = 2\mathrm{e}^{x}\sin x + c$

$\Rightarrow I = \mathrm{e}^{x}\sin x + A$

$\therefore \mathrm{e}^{x}y = \mathrm{e}^{x}\sin x + A$

$\Rightarrow y = \sin x + A\mathrm{e}^{-x}$

15) b)

As $x \to \infty$, $\mathrm{e}^{-x} \to 0$

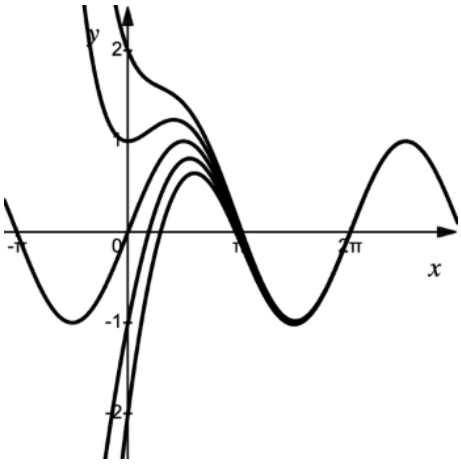
So if $A > 0$, then $y \to \sin x$

If $A = 0$, then $y = \sin x$

If $A < 0$, then $y \to \sin x$

So every solution curve approaches the curve $y = \sin x$

15) c)



16) a)

$\frac{\mathrm{d}y}{\mathrm{d}x} + y = 4\sinh x$

Integrating Factor

$= \mathrm{e}^{\int 1\mathrm{d}x}$

$= \mathrm{e}^{x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} + y = 4\sinh x$

$\Rightarrow \mathrm{e}^{x}\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{e}^{x}y = 4\mathrm{e}^{x}\sinh x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{x}y) = 4\mathrm{e}^{x}\sinh x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{x}y) = 4\mathrm{e}^{x} \times \frac{\mathrm{e}^{x} - \mathrm{e}^{-x}}{2}$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{x}y) = 2\mathrm{e}^{x}(\mathrm{e}^{x} - \mathrm{e}^{-x})$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{x}y) = 2\mathrm{e}^{2x} - 2$

$\Rightarrow \mathrm{e}^{x}y = \mathrm{e}^{2x} - 2x + A$

$\Rightarrow y = \mathrm{e}^{x} + (A - 2x)\mathrm{e}^{-x}$

16) b)

$y|_{x=0} = \mathrm{e}^0 + (A - 2 \times 0)\mathrm{e}^0$

$= 1 + A$

So the solution curves intersect the y-axis at $(0, 1 + A)$

16) c)

As $x \to \infty$, $\mathrm{e}^{-x} \to 0$ and $\mathrm{e}^x \to \infty$

So every solution curve tends to infinity as x tends to infinity

As $x \to -\infty$, $\mathrm{e}^{-x} \to \infty$ and $\mathrm{e}^x \to 0$

So every solution curve tends to infinity as x tends to negative infinity

16) d)

$y = \mathrm{e}^x + (A - 2x)\mathrm{e}^{-x}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{e}^x - (A - 2x)\mathrm{e}^{-x} - 2\mathrm{e}^{-x}$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\therefore \mathrm{e}^x - (A - 2x)\mathrm{e}^{-x} - 2\mathrm{e}^{-x} = 0$

$\Rightarrow \mathrm{e}^x - A\mathrm{e}^{-x} + 2x\mathrm{e}^{-x} - 2\mathrm{e}^{-x} = 0$

$\Rightarrow \mathrm{e}^{2x} - A + 2x - 2 = 0$

$\Rightarrow A = \mathrm{e}^{2x} + 2x - 2$

Substitute $A = \mathrm{e}^{2x} + 2x - 2$ into $y = \mathrm{e}^x + (A - 2x)\mathrm{e}^{-x}$:

$y = \mathrm{e}^x + \left((\mathrm{e}^{2x} + 2x - 2) - 2x\right)\mathrm{e}^{-x}$

$\Rightarrow y = \mathrm{e}^x + (\mathrm{e}^{2x} - 2)\mathrm{e}^{-x}$

$\Rightarrow y = \mathrm{e}^x + \mathrm{e}^x - 2\mathrm{e}^{-x}$

$\Rightarrow y = 2\mathrm{e}^x - 2\mathrm{e}^{-x}$

$\Rightarrow y = 4 \times \frac{\mathrm{e}^x - \mathrm{e}^{-x}}{2}$

$\Rightarrow y = 4 \sinh x$

Alternatively:

The original differential equation is $\frac{\mathrm{d}y}{\mathrm{d}x} + y = 4 \sinh x$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\therefore 0 + y = 4 \sinh x \Rightarrow y = 4 \sinh x$

16) e)

